2D Mechanical Clock Simulation:

A Project Proposal For

The Partial Fulfilment Of The Requirement In

Computer Graphics and Visualisation

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Project Proposal

2D Mechanical Clock Simulation

Description:

This project is a graphical application developed using **C++** and OpenGL. The core of the simulation relies on mathematical rotation matrices and trigonometry to calculate the exact positions of the clock hands, hour, minute, and second, in real-time. We present two options, one to set a custom starting time and other, to sync with the system's hardware clock. The project focuses on the precise application of trigonometry, coordinate geometry and the graphics pipeline to transform time data into a visual representation.

Methodology:

Tools and Technologies

The development of this simulation will utilize the following software tools and standards:

- **C++ Programming Language:** Used for the entire codebase, including the implementation of transformation matrices and system-level time fetching.
- **OpenGL (Open Graphics Library):** Used as the primary API for rendering 2D primitives, such as points, lines, and polygons, and managing the graphics state.
- **GLUT (OpenGL Utility Toolkit):** Utilized for window creation and managing the main event display loop.
- **Time Library (`ctime`):** A standard C++ library used to interface with the hardware clock to retrieve current hours, minutes, and seconds.
- **Math Library (`cmath`):** Used to perform trigonometric calculations required for circular geometry and rotation transformations.

Implementation Phases

1. **Rasterization:** The clock components are defined using Cartesian coordinates. The clock face is constructed by calculating points along a circular path using **Mid-point circle logic**, while the hands and markers are drawn using line drawing algorithms (Bresenham)

Mathematically, If (xc, yc) represents the centre of the circle, then (x,y) represents the other end of a hand of length L, at a degree θ:
$x = xc + L \cdot \sin(\theta)$
$y = yc - L \cdot \cos(\theta)$
1 hour = 30 degrees.
1 minute = 6 degrees.
1 second = 6 degrees.

2. **2D Transformation Engine:** Rotation logic is implemented to manipulate the clock hands.
    - **Rotation:** Calculating the precise angle for the second, minute, and hour hands relative to the center (0,0). Every 1 second, all the hands are updated by apply rotation matrix on all three hands' coordinates through fixed angles.
    - **Static Scaling & Translation:** Setting a fixed scale and position for the clock within the viewport to ensure it is perfectly centered and visible upon execution.
3. **Real-Time Animation Logic:** The application uses a timer callback function (e.g., `glutTimerFunc`) to refresh the screen at regular intervals. During each refresh, the program updates the rotation angles of the hands to create a continuous and accurate "sweeping" animation.

Simplified Algorithm:

1. Get initial hand degrees based on initial time.
2. Get endpoints of each hand based on the degrees.
3. Get all the points using line drawing algorithms.
4. Plot the initial hands.
5. Delay 1 second.
6. Apply rotation matrix on all the hands separately to get points of the new hand position.
7. Clear the frame buffer, redraw the static clock face, and plot the updated hand positions.
8. Jump to 5.

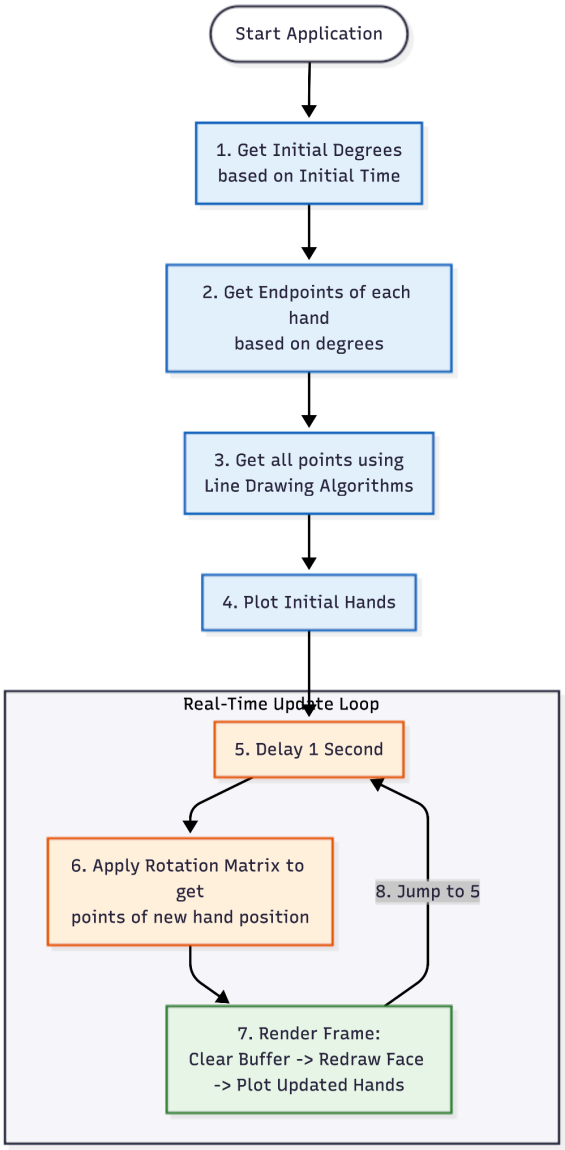

Fig: Block Diagram:

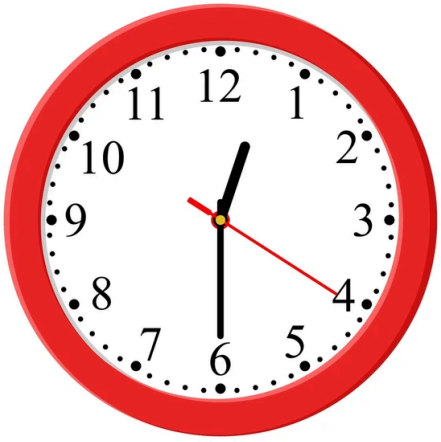

Fig: Expected Output:

Team Members

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